

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Built MCP LLM pipelines (Claude, GPT-4o, Gemini); document processing (PDF gen, S3, email); Playwright coverage. - Owned 7-deployment Node.js backend: PostgreSQL, Redis, Auth0 RBAC, Temporal.io, Docker, Terraform, Sentry.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Cloud-based genomic platform on Google Cloud Run with JBrowse 2; 100+ researchers, 500GB datasets worldwide. - Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI-driven workflows including health claim automation using OCR, NLP, and ML; improved accuracy by 25%. - Automated deployments on AWS and GCP, integrated Grafana dashboards, led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for data-driven system automating support ticket analysis, reducing processing time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Demand Paging – Xinu OS	Systems Programming	Oct 2024 – Dec 2024
- Engineered 8-store virtual memory system in C for Xinu OS with Second-Chance page replacement; cut page faults 25%. - Added 7+ kernel system calls, increasing memory performance by 15% via inverted page tables and custom fault handler.		
PulseRank	Real-Time Leaderboard	2026
- TypeScript/NestJS leaderboard: MySQL outbox, MongoDB TTL, Redis sorted sets; Testcontainers integration tests. - Shipped OpenTelemetry + Prometheus on all hot paths; load-tested with k6; deployed via GitHub Actions CI/CD.		
VectorLift	Semantic Search Engine	2026
- Semantic search in Python on MS MARCO: BM25, dense bi-encoder (in-batch negatives), cross-encoder re-ranker. - Scored NDCG@10, MRR@10, MAP; FastAPI serving with per-stage latency; Kafka-driven Qdrant corpus indexing.		
Inventory Reservation Service	Backend API	2026
- Java 21/Spring Boot REST API: PostgreSQL row-level locking, idempotent POSTs, RFC 9457 errors, scheduled expiry. - Testcontainers: two concurrent POSTs produce exactly one 200, one 409; Flyway, Docker, GitHub Actions CI.		

Languages and Technologies

- Languages:** C, Python, Java, TypeScript, JavaScript, SQL, Bash
- Systems & Distributed:** Kernel programming, virtual memory, Kafka, Spark, Hadoop, Docker, Kubernetes, Terraform
- Backend & Cloud:** Spring Boot, Node.js, FastAPI, NestJS, PostgreSQL, Redis, MySQL, AWS, GCP, GitHub Actions
- AI & Tools:** Claude/GPT-4o/Gemini APIs, MCP, LangChain, OpenTelemetry, Prometheus, Testcontainers

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; developed 3 complete MVPs within 48-hour builds.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.